The Semiconductor Diode

Mihir Bhatt

Rocky Wang

Faculty of Engineering and Architectural Science
Toronto Metropolitan University
Toronto, Canada

PCS224-06 Solid State Physics

October 2026

Introduction:

A semiconductor diode is made by joining two n-type and p-type semiconductors to create a p-n junction, which acts as a one-way valve for electric current to flow through it. When a voltage source is connected to the diode, a large current will flow through the junction if it is forward-biased. In contrast, almost no current will flow through the pn junction if the current is reverse-biased. The exponential relationship between the current flowing through the semiconductor diode and the applied voltage is based on the thermal voltage, which is dependent on the temperature of the diode at that specific time period and Boltzmann's constant. The purpose of this experiment is to use an n-p-n silicon junction to obtain an ideal diode under three conditions: room temperature, an ice water bath (low-freezing temperature), and hot water (high temperature). By measuring the collector current through the base-collector junction and the voltage across the base-emitter junction, this experiment aims to derive a linear relationship between the current and voltage, calculate the reverse saturation current, and experimentally determine Boltzmann's constant value.

Theory:

A semiconductor diode consists of a p-doped semiconductor that is joined to an n-doped semiconductor. This acts as sort of a one way valve that allows current to easily flow through in one direction while limiting the flow in the other direction. When the p side has a higher potential than the n side, the diode is forward biased and a large current flows through. If the polarity is reversed, the diode is reversed biased and almost no current flows. The current-voltage relationship of an ideal pn junction is given by the Shockley diode equation:

$$I = I_0[exp(\frac{V}{V_t}) - 1] \qquad (1)$$

Where I is the current through the junction, V is the applied voltage, I_0 is the maximum current that can flow under reverse bias (reverse saturation current) and V_t is the thermal voltage. The thermal voltage is modeled by the equation:

$$V_t = \frac{k_B T}{|q|} \qquad (2)$$

Where k_B is Boltzmann's constant, T is the temperature and $|q|$ is the magnitude of the charge carried by each carrier. Equation (1) models both bias conditions where under strong forward bias ($V \gg V_t$), the exponential term is much larger than 1, so the -1 can be neglected and the current grows exponentially with voltage. Under strong reverse bias ($V < 0$ and $|V| \gg V_t$), the exponential term approaches zero and the current levels off at $I \approx -I_0$. At $V = 0$, no current flows. Although a reverse saturation current still flows under reverse bias, since I_0 is extremely small compared to what the forward biased current is, in most scenarios, it is reasonable to say that the reverse current is negligible and is approximately zero.

For the forward-bias voltages used in this experiment, $V \gg V$ so Equation (1) reduces to:

$$I \approx I_0 \, exp(\frac{|q|V}{k_B T}) \qquad (3)$$

Taking the natural logarithm of both sides gives:

$$ln\,(I) \;=\; \frac{|q|}{k_B T} V \;+\; ln\,(I_0) \qquad (4)$$

This gives a linear relationship where when $ln\, I$ is plotted against V at a constant temperature, the result is a line with a slope of $m = \frac{|q|}{k_B T}$ and an intercept of $b = ln\, I_0$. Boltzmann's constant and reverse saturation current can therefore be calculated from the line using the equations:

$$k_B = \frac{|q|}{mT} \qquad (5)$$

$$I_0 = e^b \qquad (6)$$

Since the slope is inversely proportional to the temperature, data at a higher temperature should give a smaller slope.

Procedure:

The Thermometer was turned on and set to observe the room temperature in degrees Celsius. Next, the multimeter was turned on and set to its diode setting, and the base and emitter wires of the n-p-n junction were connected to the multimeter, where the base was connected to the positive terminal and the emitter was connected to the ground (negative) terminal of the multimeter. The forward bias was observed on the multimeter and recorded in a table. The base and emitter wires were then switched to observe the reverse bias on the multimeter. The observations were recorded in a table. The current supply was then turned on and connected to the n-p-n junction based on the diagram given in the lab, ensuring that the base lead was connected directly to the collector lead. The temperature on the thermometer was recorded before starting to vary the currents on the power supply. The current was then set to the lowest reading of 1.4 mA and was doubled each time until a reading of 179.2 mA was reached. The voltage reading observed on the multimeter at each reading was recorded in a table. Once the last reading was recorded, the final temperature reading on the thermometer was recorded for further calculations. The transistor was then submerged into a cup of ice water along with the temperature probe. The setup was then rested for approximately a minute for the diode and thermometer to reach a consistent temperature reading. The reading on the thermometer was recorded and the current was varied again starting from 1.4 mA to 179.2 mA, taking the reading across the multimeter for each current observation. Once all voltage readings were recorded, the final temperature of the ice water was recorded using the thermometer. The ice water was then drained and switched out for hot water. The semiconductor diode and thermometer were once again placed inside the hot water and rested for approximately a minute for the thermometer reading to reach equilibrium. The initial temperature reading was recorded, and the current was once again varied from 1.4 mA to 179.2 mA, taking the voltage readings across the multimeter at each current variation. The final temperature reading across the thermometer was recorded once all readings were taken. The power supply and thermometer were powered off, the water was drained, and all wires were unplugged.

Results and Calculations:

Table 1: Multimeter Diode Test Readings

Junction Under Testing	**Forward Bias Multimeter Reading**(mV)	**Reverse Bias Multimeter Reading** (mV)
Base - Emitter	489	Overload
Base - Collector	484	Overload

Table 2: Recorded Temperature Measurements at Room Temperature, the Temperature of Ice Water, and the Temperature of Hot Water

Case:	**Starting Measured Temperature** $(°C)$	**Final Measured Temperature** $(°C)$	**Uncertainty** δT $(°C)$	**Starting Measured Temperature in Kelvin** (K)	**Final Measured Temperature in Kelvin** (K)
Room Temperature	20.8	22.2	$\pm$ 0.7	293.75	295.75
Ice Water Temperature	9.1	8.8	$\pm$ 0.15	282.35	281.95
Hot Water Temperature	78.4	74.6	$\pm$ 1.9	351.55	347.75

Table 3: Recorded Observations of the Collector Current in mA and the Base-Emitter Voltages of the Semiconductor Diode at Each Temperature

Collector Current / Power Supply Current I (mA)	**B - E Voltage (**V_{RT}**) at Room Temperature** (V)	**B - E Voltage (**V_{IW}**) at Ice Water Temperature** (V)	**B - E Voltage (**V_{HW}**) at Hot Water Temperature** (V)
1.4	0.503	0.536	0.361
2.8	0.522	0.555	0.385
5.6	0.541	0.573	0.408
11.2	0.561	0.593	0.434
22.4	0.582	0.613	0.461
44.8	0.605	0.635	0.491
89.6	0.630	0.660	0.526
179.2	0.658	0.689	0.565

Table 4: Measured Uncertainty Values for The Current and Voltage Readings

Uncertainty $\delta I\ (mA)$	**Uncertainty** $\delta V_{RT}\ (V)$	**Uncertainty** $\delta V_{IW}\ (V)$	**Uncertainty** $\delta V_{BW}\ (V)$
$\pm\ 0.05$	$\pm\ 0.0005$	$\pm\ 0.0005$	$\pm\ 0.0005$

1) Assuming that V ≫ Vt, the diode equation becomes approximately:

$$I\ =\ I_0 exp[\frac{|q|V}{k_b T}] \qquad (3)$$

Using this equation as a guide, plot your three data sets in such a way that each one is a straight line.

Taking the logarithm of both sides of equation (3):

$$ln(I)\ =\ ln(I_0 exp[\frac{|q|V}{k_b T}])$$

$$=\ ln(I)\ =\ ln(I_0)\ +\ ln(exp[\frac{|q|V}{k_b T}])$$

$$=\ ln(I)\ =\ ln(I_0)\ +\ [\frac{|q|V}{k_b T}]ln(e)$$

$$=\ ln(I)\ =\ ln(I_0)\ +\ [\frac{|q|V}{k_b T}]$$

$$=\ ln(I)\ =\ (\frac{|q|}{k_b T})V\ +\ ln(I_0) \qquad (4)$$

Therefore, the linearized equation of equation (3) is described by equation (4), where V is the independent variable in Volts, $ln(I)$ is the dependent variable in Amperes, $\frac{|q|}{k_b T}$ is the slope of the function, and $ln(I_0)$ is the y-intercept in Amperes.

2) Fit your data with straight lines. Use the slope and intercept of each line to determine Boltzmann's constant and the reverse saturation current I_0.

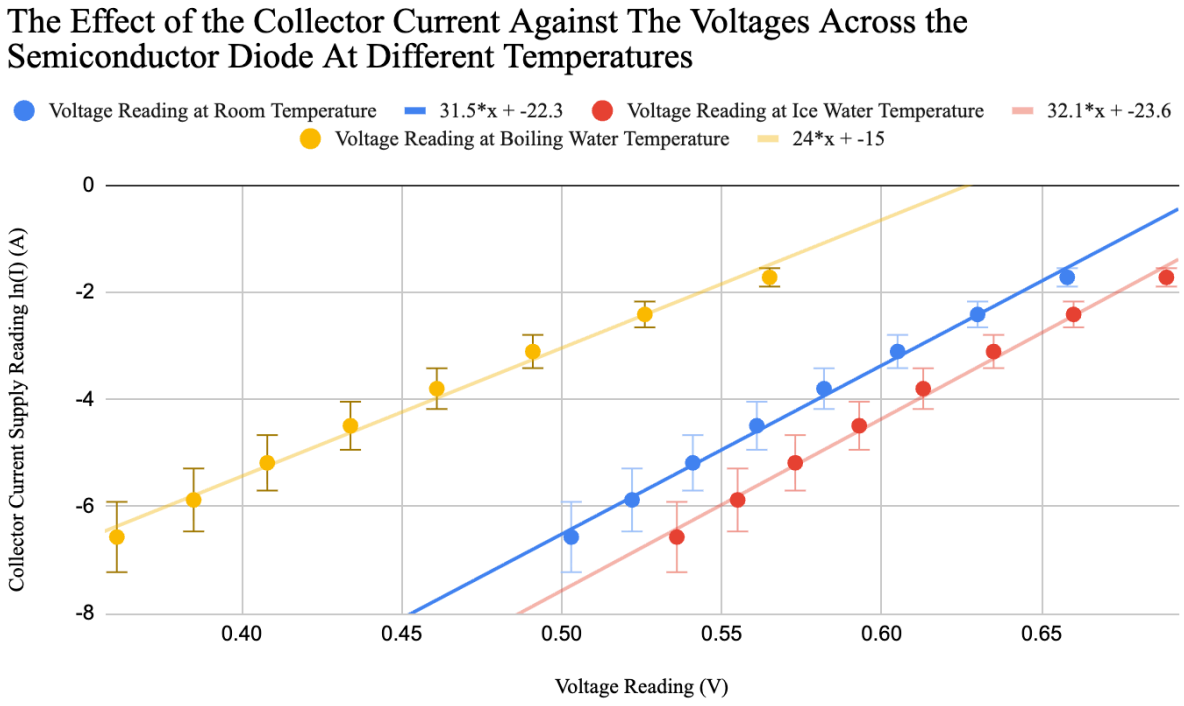


Figure 1: The Effect of the Collector Current Against the Voltages Across the Semiconductor Diode at Different Temperatures: This graph shows the linearized relationship between the base-emitter voltages in Volts and the natural logarithm of the collector currents in Amperes.

To calculate Boltzmann's constant at each observed temperature, the slope of the graph, m, can be rearranged as seen in equation (4), where $m = \frac{|q|}{k_b T}$. The reverse saturation currents can be found similarly from each linear relationship by rearranging the y-intercept of the graph as found in equation (4), where $b = ln(I_0)$. The Boltzmann's constants for each temperature case can be solved as shown below:

Case 1: Finding Boltzmann's Constant and Reverse Saturation Current at Room Temperature

$$T_{avg} = \frac{T_{start} + T_{finish}}{2} = \frac{293.75\ K + 295.75\ K}{2} = 294.75\ K$$

$$k_b = \frac{|1.602 \times 10^{-19}|}{31.5(294.75)} = 1.725 \times 10^{-23} \frac{J}{K}$$

$$I_0 = e^{-22.28} = 2.11 \times 10^{-10} A$$

Case 2: Finding Boltzmann's Constant and Reverse Saturation Current at Ice Water Temperature

$$T_{avg} = \frac{T_{start} + T_{finish}}{2} = \frac{282.35\ K + 281.95\ K}{2} = 282.15\ K$$

$$m = \frac{|q|}{k_b T}$$

$$k_b = \frac{|q|}{mT}$$

$$k_b = \frac{|1.602\times10^{-19}|}{32.1(282.15)} = 1.769 \times 10^{-23}\frac{J}{K}$$

$$b = ln(I_0)$$

$$e^b = e^{ln(I_0)} = I_0$$

$$I_0 = e^{-23.63} = 5.48 \times 10^{-11} A$$

Case 3: Finding Boltzmann's Constant and Reverse Saturation Current at Hot Water Temperature

$$T_{avg} = \frac{T_{start}+T_{finish}}{2} = \frac{351.55\ K + 347.75\ K}{2} = 349.65\ K$$

$$k_b = \frac{|1.602\times10^{-19}|}{24(349.65)} = 1.909 \times 10^{-23}\frac{J}{K}$$

$$I_0 = e^{-15.02} = 3.00 \times 10^{-7} A$$

Table 5: Calculated Boltzmann's Constant and Reverse Saturation Current Values Derived From The Observations Recorded Within the Experiment

Case:	**Boltzmann's Constant Value, k_b ($\frac{J}{K}$)**	**Reverse Saturation Current, I_0 (A)**
Room Temperature	1.725×10^{-23}	2.11×10^{-10}
Ice Water Temperature	1.769×10^{-23}	5.48×10^{-11}
Hot Water Temperature	1.909×10^{-23}	3.00×10^{-7}

3) You should now have three measurements of k_b and I_0. Use these measurements to quote a final result for k_b and for I_0 including uncertainty. (Remember that if you have several measurements of the same quantity, a good estimate for the uncertainty in this quantity is the standard deviation of the measurements).

The standard mean equation is given by equation (7) and the equation to determine the standard deviation is given by equation (8), as shown below:

$$mean = \overline{X} = \frac{x_1+x_2+x_3}{3} \qquad (7)$$

$$Deviation = \delta x = \sqrt{\frac{(x_1-\overline{X})^2+(x_2-\overline{X})^2+(x_3-\overline{X})^2}{3-1}} \qquad (8)$$

Plugging in the required values into equations (7) and (8) yields:

$$\overline{k_b} = \frac{1.725+1.769+1.909}{3} \times 10^{-23} = 1.801 \times 10^{-23} \frac{J}{K}$$

$$\delta k_b = \sqrt{\frac{(1.725-1.801)^2+(1.769-1.801)^2+(1.909-1.801)^2}{3-1}} \times 10^{-23} = 0.10 \times 10^{-23} \frac{J}{K}$$

Unlike Boltzmann's constant, the reverse saturation current is not a fixed constant. I_0 depends on the number of thermally generated charge carriers in the semiconductor, which increases with temperature. Since only one value of I_0 was obtained at each temperature, a standard deviation cannot be used for its uncertainty. Instead, the uncertainty was found from the standard error of the fitted y-intercept, δb. Since $I_0 = e^b$, the uncertainty propagates as:

$$\delta I_0 = I_0 \cdot \delta b \qquad (9)$$

Applying Equation (9) with the intercept standard errors from the linear fit gives the values in Table 6:

Table 6: Reverse Saturation Current at Each Temperature

Case:	**Average Temperature** (K)	**Intercept** $b = ln(I_0)$	**Reverse Saturation Current** I_0 (A)
Ice Water Temperature	282.15	-23.63 ± 0.63	$(5.5 \pm 3.5) \times 10^{-11}$
Room Temperature	294.75	-22.28 ± 0.52	$(2.1 \pm 1.1) \times 10^{-10}$
Hot Water Temperature	349.65	-15.02 ± 0.42	$(3.0 \pm 1.3) \times 10^{-7}$

The uncertainties of I_0 are relatively large because the intercept is found by extrapolating each line back to $V = 0$ which is far outside the measured voltage range of 0.36–0.69 V. Small errors in the slope therefore produce large errors in the intercept, and exponentiating to find I_0 amplifies these errors further.

Therefore, the final experimental value of Boltzmann's constant is $k_b = (1.80 \pm 0.10) \times 10^{-23} \frac{J}{K}$. The reverse saturation current increases with temperature, from $(5.5 \pm 3.5) \times 10^{-11}\ A$ at 282.15 K to $(2.1 \pm 1.1) \times 10^{-10}\ A$ at 294.75 K and $(3.0 \pm 1.3) \times 10^{-7}\ A$ at 349.65 K.

4) Compare your results for kB with the accepted value, including uncertainty. Are the results consistent? If there is a disagreement with the accepted value, discuss why this may be the case.

Table 7: The Experimentally Calculated and Accepted Values of Boltzmann's Constant

	Boltzmann's Constant, k_b
Experimentally Calculated Value	$(1.80 \pm 0.10) \times 10^{-23} \frac{J}{K}$
Accepted Value	$1.381 \times 10^{-23} \frac{J}{K}$

Calculating Percent Error of Boltzmann's Constant:

$$Percent\ Error = \frac{|k_{experiment} - k_{accepted}|}{k_{accepted}} \times 100\% = \frac{|1.801 - 1.381|}{1.381} \times 100\% = 30.4\%$$

Based on the results observed within the lab and the calculated Boltzmann's constant, the results are not statistically consistent with the theoretically accepted value. The accepted value of $1.381 \times 10^{-23} \frac{J}{K}$ lies outside of the experimental calculations, which yielded a value of $(1.80 \pm 0.10) \times 10^{-23} \frac{J}{K}$. A reason as to why there is an experimental discrepancy between the experimental and accepted values may be due to the omission of the diode ideality factor n. From equation (3) in this experiment, the equation assumes an ideality factor of n = 1. In real silicon n-p-n diodes, the ideality factor can vary between 1 and 2. This means that the revised equation of equation (3) is: $I = I_0 exp[\frac{|q|V}{nk_bT}]$, where n is the newly observed ideality factor. This means the calculated Boltzmann's constant is actually yielding a value of nk_b. From rearranging the equation of $k_{b,experimental} = nk_{b,accepted}$, the ideality factor is n = 1.80 / 1.381 ≈ 1.30, which explains the 30.4% discrepancy between the experimental and accepted Boltzmann's constant values.

Discussion and Conclusions:

The multimeter diode test confirmed that both the base-emitter and base-collector junctions of the n-p-n transistor only conduct when forward-biased. Each junction gave a forward reading of approximately 0.49 V and an overload reading when reversed (Table 1). This confirms that the transistor consists of two pn junctions and that each acts as a one-way valve for current. Under forward bias, the collector current increased exponentially with the base-emitter voltage. Every time the current doubled, roughly a constant increase in voltage of about 20 mV was required. Plotting $ln(I)$ against V produced a straight line at all three temperatures, with $R^2 > 0.99$ in each case (Figure 1). This is consistent with the linearized diode equation (4). The slope also decreased with increasing temperature, from 32.1 V^{-1} in ice water to 31.5 V^{-1} at room temperature and 24.0 V^{-1} in hot water, as predicted by $m = \frac{|q|}{k_b T}$.

As discussed earlier, the experimental value of $k_b = (1.80 \pm 0.10) \times 10^{-23} \frac{J}{K}$ is 30.4% above the accepted value, which corresponds to an ideality factor of approximately 1.30. The measurement uncertainties in current (±0.05 mA) and voltage (±0.0005 V) are far too small to account for a deviation of this size. In addition, all three temperature cases overestimate k_b. This indicates that the error isn't just random and that several experimental factors likely pushed the effective ideality factor above 1. The most significant of these factors is likely series resistance and self-heating at high currents. The plots of $ln(I)$ against V curve slightly downward as current increases. Fitting only the four lowest-current points at room temperature gives a slope of 35.9 V^{-1} which corresponds to an ideality factor of about 1.10, while the four highest-current points give a slope of only 27.3 V^{-1}. At currents near 180 mA, part of the measured voltage drops across the resistance of the leads, cable and bulk silicon in the transistor along rather than just across the junction itself. The power dissipated in the transistor at these currents also heats the junction above the measured water temperature. Both effects reduce the slope of the fitted line and inflate the calculated value of k_b. A second source of error is the mismatch between the measured temperature and the actual junction temperature. The temperature probe measured the water rather than the junction directly, and with a minute of immersion, the transistor may not have fully reached thermal equilibrium with the bath. The hot water also cooled by 3.8 K over the course of the measurements, so the junction temperature changed while data was being collected. Averaging the starting and final temperatures only partly accounts for this drift, which likely explains why the hot water case produced the largest error, with $k_b = 1.909 \times 10^{-23} \frac{J}{K}$. Finally, there may still be some recombination current that has contributed to the measured current. Measuring the collector current excludes most of the recombination current, since it exits through the base terminal, but this separation is not perfect. Any recombination current that is still measured follows $exp(\frac{|q|V}{2k_b T})$ rather than the ideal exponential, which pushes the effective ideality factor toward 2 and further increases the calculated value of k_b. The accuracy of this experiment could be improved by limiting measurements to lower currents (below about 20 mA) to reduce series resistance and self-heating, giving the transistor more time to reach thermal equilibrium in each bath, using a temperature-controlled bath so the temperature does not drift during measurements and measuring the transistor's case temperature directly to get a better estimate of the junction temperature.

This experiment confirmed the current–voltage behaviour of a pn junction described by the ideal diode equation. The transistor's junctions conducted only under forward bias, and the collector current increased exponentially with the base–emitter voltage, producing linear plots of $ln(I)$ against V at all three temperatures. From the slopes, Boltzmann's constant was determined to be $(1.80 \pm 0.10) \times 10^{-23} \frac{J}{K}$, which is 30.4% higher than the accepted value of $1.381 \times 10^{-23} \frac{J}{K}$ and not consistent within uncertainty. This discrepancy is attributed mainly to non-ideal diode behaviour (n ≈ 1.30), caused by series resistance and self-heating at high currents and by imperfect thermal equilibrium between the transistor and its bath. The reverse saturation current increased strongly with temperature, from 5.48×10^{-11} A at 282.15 K to 3.00×10^{-7} A at 349.65 K, which explains why diode current increases with temperature at a fixed forward voltage.

References:

[1] D. A. Neamen, *Semiconductor Physics and Devices: Basic Principles*, 4th ed. New York, NY, USA: McGraw-Hill, 2012.

[2] Toronto Metropolitan University, "The Semiconductor Diode," *PCS224 Solid State Physics Laboratory Manual*, Toronto, ON, Canada, 2026.

[3] C. Honsberg and S. Bowden, "Ideality Factor and I0," *PVEducation*. Accessed: Oct. 23, 2026. [Online]. Available: https://www.pveducation.org/pvcdrom/idealty-factor-and-i0

[4] National Institute of Standards and Technology, "CODATA Values of the Fundamental Physical Constants," *NIST Reference on Constants, Units, and Uncertainty*. Accessed: Oct. 23, 2026. [Online]. Available: https://physics.nist.gov/cuu/Constants/

Appendix:

1. As mentioned in the theory section, under reverse bias, Equation (1) shows that the current does not drop to exactly zero but levels off at $I \approx -I_0$, the reverse saturation current. This current is carried by the small number of minority carriers that are generated from thermal energy near the junction and swept across it by the built-in electric field. However, I_0 is extremely small. In this experiment it was found to be on the order of 10^{-10} A at room temperature, while the forward-bias currents measured were on the order of milliamperes, which is more than 10^8 times larger. Because the reverse current is so many orders of magnitude smaller than the forward current, it has a negligible effect in most practical circuits. Therefore it is reasonable to say that a diode allows no current to flow under reverse bias. This was also observed in the multimeter

diode test where both junctions of the transistor gave an overload reading when reverse biased, indicating no measurable conduction (Table 1).

2. Using Equation (2), the thermal voltage is approximately 24.3 mV at the ice water temperature (282.15 K), 25.4 mV at room temperature (294.75 K), and 30.1 mV at the hot water temperature (349.65 K). The smallest voltages recorded in each data set were 0.536 V, 0.503 V, and 0.361 V respectively (Table 3). These are 22.0, 19.8, and 12.0 times the corresponding thermal voltages. The assumption $V \gg V_t$ is therefore valid for every data point collected, and it does not contribute to the discrepancy between the experimental and accepted values of k_b.
3. If the temperature is raised while the forward bias voltage stays fixed, the current through the diode increases. In theory however, equation (3) alone might suggest the opposite, since a higher T reduces the exponent $\frac{|q|V}{k_b T}$. What this does not consider, is that the reverse saturation current I_0 depends on the number of thermally generated carriers in the semiconductor, which grows very rapidly with temperature. This increase in I_0 far outweighs the smaller exponent, so the net current rises. The results support this: I_0 increased from $5.48 \times 10^{-11}\ A$ in ice water to $2.11 \times 10^{-10}\ A$ at room temperature and $3.00 \times 10^{-7}\ A$ in hot water (Table 6). That is an increase of nearly four orders of magnitude over a 68 K range. Equivalently, the same current was reached at a lower voltage when the transistor was hotter. At 1.4 mA, the base–emitter voltage dropped from 0.536 V in ice water to 0.361 V in hot water (Table 3), approximately −2.6 mV/K.